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MILITARY SPECIFICATION

TAMPER, RAM TYPE, GENERAL BACKFILL, GASOLINE ENGINE POWERED, HAND OPERATED, COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial, hand operated, gasoline-engine-powered, ram type, general backfill, soil tamper.

1.2 Classification.

Type I - 1000 pounds of applied force (minimum).
Type II - 1400 pounds of applied force (minimum).

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 3895

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Material.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-C-12044	- Chests, Plywood.
MIL-P-19834	- Plates, Identification or Instruction, Metal Foil, Adhesive backed General Specification For.
MIL-T-22085	- Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Material.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems.

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2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

OSHA Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF AGRICULTURE (USDA)

Forest Service Standard 5100-1A

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

SAE J335 - Multiposition Small Engine Exhaust System.
SAE J905 - Fuel Filter Test Method.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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CONSTRUCTION INDUSTRY MANUFACTURERS ASSOCIATION (CIMA)

Light Equipment Manufacturers Bureau - Standard No.1.

(Application for copies should be addressed to the Construction Industry Manufacturers Association, 111 E. Wisconsin Ave., Milwaukee, WI 53202.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The tamper shall be a hand operated and controlled, gasoline-engine powered, ram type soil compactor that is used for general backfill compaction. The tamper shall be the current commercial production model of a standard product and except for product improvements which have been incorporated in the commercial production units, shall have been marketed and in commercial field use in the exact configuration with all components to be furnished for at least one year prior to opening of the technical proposal or bid. The tamper shall have proven widespread commercial usage experience and acceptance in most types of geographic areas, climatic conditions and altitudes throughout the continental United States. The tamper shall be equipped with all components necessary to enable the tamper to function reliably and efficiently in a sustained operation. The tamper shall conform to all applicable Federal laws and regulations governing safety, noise levels and pollution which are in effect on date of bid.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of tampers from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish tampers conforming to this specification.

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3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The tamper shall be fabricated from compatible materials inherently corrosion resistant, or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponents, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the tamper may be newly fabricated from recovered materials to the maximum extent practicable, provided the tamper produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the tampers.

3.4 Engine. The engine type shall be a 2-cycle, air-cooled, gasoline type.

3.4.1 Starter. The engine shall be equipped with an automatic rewind type starter.

3.4.2 Fuel filter. The engine shall be equipped with a fuel filter conforming to SAE J905 which will prevent dirt and other foreign material from entering the carburetor and combustion chamber.

3.4.3 Air filter. The intake air shall be filtered to prevent dust and dirt from entering the engine.

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3.4.4 Muffler. The engine shall be equipped with an exhaust muffler reducing noise levels conforming to SAE J335 or USDA Forest Service Standard 5100-1A.

3.4.5 Fuel tank. The fuel tank shall be cast or molded non-corrosive type material. Construction and mounting of the fuel tank shall be that neither vibration nor excessive tightening of the mounting provisions shall cause leaks. The fill-opening cap shall have a leakproof vent to enable positioning of the tamper from the vertical to the horizontal in all positions without leakage. The fuel tank capacity shall conform to the requirements specified in 3.13d. The manufacturer's recommended mixture of gasoline and oil (ratio) shall be on an instruction plate or permanently marked on the tamper or cap.

3.5 Tamping or ramming shoe. The tamper shall have a ramming shoe that is rectangular in configuration and has a minimum surface area of 125 square inches. The surface area is defined as the total vertical projected area of the shoe including all bends and radii. The shoe shall consist of wear-resistant type material. Provision shall be made to restrict in-service turning. The effective shoe contact area, as defined by CIMA-LEMB standard No. 1, shall not be less than 95 square inches.

3.5.1 Accessories. When specified (see 6.2), one each of the following accessories shall be furnished with each tamper:

a. A 6-inch width shoe with 12-inch extension (nominal).
b. A 4-inch width shoe with 12-inch extension (nominal).

3.6 Handles. The tamper shall be equipped with handles that serve three purposes: (1) operating, (2) lifting or carrying and (3) engine component protection. The handles shall be:

a. Mounted with a shock absorbing means to minimize operator fatigue from vibration.
b. Capable of being used to lift or carry the tamper into an out of confined areas and loading and unloading onto and from the transporter.
c. Designed to prevent damage to the engine and its components during compaction operation and transportation between job sites.
d. To allow for three people using two hands each to carry an evenly distributed load.

3.7 Drive system. The drive system shall be gear and spring driven. The gear and spring shall be oil lubricated, totally enclosed, and dust and water proof.

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3.8 Compacting system. The tamper shall have shock absorbing means incorporated into the compacting system to minimize the impact shock that is transferred into the machine. Provision shall be made to prevent the functioning of the compaction system while the engine is being started, idled, or adjusted.

3.9 Lubricants. The procedure for the selection of lubricants shall conform to MIL-STD-838, section 5. Lubricants selected shall conform to MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products Lists (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.10 Special tools. Any special tools required for operator maintenance of the tamper shall be provided with each tamper.

3.11 Carrying case. Unless otherwise specified (see 6.2), a reusable carrying case (chest) with skids, conforming to MIL-C-12044 shall be provided for each tamper.

3.11.1 Size. The interior of the chest shall provide a minimum of one inch clearance between all chest panels and adjacent extreme surfaces of the tamper.

3.11.2 Blocking. Blocking and tiedown features to prevent movement of the tamper in longitudinal, lateral and vertical directions shall be incorporated into the chest and secured conforming to MIL-C-12044. These features shall be positioned in such a manner to insure that the tamper can only be placed in the carrying case in the proper position and shall not cause damage to any parts of the tamper.

3.11.3 Paint. Unless otherwise specified (see 6.2), the final paint color shall conform to the requirements specified in 3.19.

3.11.4 Marking. Unless otherwise specified (see 6.2), marking on the outside of the chest lid shall be as indicated below. The letters "US Army" to be two inches high and the remaining letters to be 0.75 inch high. All letters shall be painted black conforming to MIL-C-53039 or MIL-C-46168.

"US ARMY
TAMPER, RAM TYPE, GENERAL BACKFILL, GASOLINE
ENGINE POWERED, HAND OPERATED"

3.11.5 Loading plan. Unless otherwise specified (see 6.2), a loading plan conforming to MIL-C-12044 shall be furnished.

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3.11.6 Publication storage. When a carrying case is required, the manufacturer shall furnish fasteners and permanently install, within each carrying case, a publications container. All publications required for each tamper shall be put in the container prior to shipment. When a carrying case is not required, publications shall be overpacked with each unit.

3.12 Maintenance characteristics. Unless otherwise specified (see 6.2), the repair, replacement, periodic inspection and servicing of components and assemblies that require periodic maintenance shall be accomplished without the need for special tools or special equipment. Special tools are defined as tools that are not available to the user of this equipment. When special tools are permitted they shall be overpacked with each unit.

3.13 Performance. The tamper shall be capable of performing the following:

a. Effectively compacting in confined areas, around footings, around abutments and retaining walls, in bomb craters, in culverts and trenches, bridge approaches and bases for concrete slabs.
b. Sustained operation in a climate where ambient temperatures reach 110°F. Sustained operation is defined as continuous operation for eight hours including downtime for normal servicing and refueling.
c. Sustained operation in an environment condition of 100 percent relative humidity. Sustained operation is defined as continuous operation for eight hours including downtime for normal servicing and refueling.
d. Continuously operate as a work tool for at least one hour at full engine throttle without refueling.
e. Compacting at a frequency of not less than 500 blows per minute at full-load-governed engine speed.
f. Developing a minimum of 1000 (type I) or 1400 (type II) pounds of applied force to the ground at full-load-governed engine speed, when measured according to the Construction Industry Manufacturers Association, Light Equipment Manufacturers Bureau Standard Number 1.
g. Operable by one operator.

3.14 Noise limits. The noise produced by the tamper shall conform to MIL-STD-1474 requirements, with the exception of MIL-STD-1474 5.2, 5.3 and 5.4, when tested to conform to 4.5.2.9. The provision of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the

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procuring activity. Hazard signs shall conform to MIL-STD-1474, 4.3, except the sign will state "HEARING PROTECTION REQUIRED WITHIN (__) FEET." The sign shall be readable at the distance indicated on the sign.

3.15 Health and safety. The tamper shall comply with all applicable OSHA standards that are in effect on the date of the opening of the bids. All rotating or reciprocating parts and all parts subject to high operating temperatures or voltage shock, that are of such a nature, or are located to be a hazard to operating or maintenance personnel shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.15.1 Safety warning label. A safety warning label shall be affixed permanently to the inside of the carrying case lid and to the fuel tank adjacent to the filler cap. This label shall read: "WARNING: DRAIN GAS TANK AND CLOSE TANK CAP VENT BEFORE PUTTING TAMPER INTO CARRYING CASE." The label shall have a yellow background with black letters and be of corrosion-resistant type material. Standard operating procedures in the instruction manuals for this item shall also contain information regarding this possible safety hazard.

3.16 Technical publications. Technical publications (commercial manuals) shall be furnished as specified in the contract or purchase order (see 6.4).

3.17 Weight. The weight of the tamper, fully serviced, less fuel, shall not be greater than 140 pounds. The weight of the tamper and the words "THREE PERSON LIFT" shall be prominently labeled on the tamper.

3.18 Dimensions. The tamper dimensions including accessories shall not exceed the following:

Length - 39.00 inches.
Width - 22.00 inches.
Height - 46.00 inches.

3.19 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the tamper normally painted shall be cleaned, treated, primed and painted conforming to MIL-T-704, type F. The finishing coat shall conform to MIL-C-46168 for camouflage green 383. When specified (see 6.2), surfaces normally painted in good commercial practice shall be cleaned, treated and painted as specified herein. The color of the finish coat shall be as specified

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(see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current process and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.20 Identification and instruction plates. The contractor shall provide all identification markings, instruction plates, or stenciled instructions normally furnished on his commercial tampers. The plates shall include the contract number, serial number, national stock number, model number and date of manufacture in addition to the instructions normally furnished. Instruction and identification plates shall conform to MIL-P-19834, type II. Adhesives or decals shall not be used to attach labels.

3.21 Workmanship. All production processes used by the contractor and his suppliers shall conform to the contractor's production drawings and manufacturing tolerances.

3.21.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.21.2 Bolted connections. Boltholes shall have the burrs removed. Washers or lockwashers shall be provided, where necessary. Matching thread areas securing bolts, or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all section 3 and 5 requirements. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined and tested conforming to referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article subjected to the tests marked "X" in column 1 of Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall conform to MIL-STD-105.

4.4.2 Examination. Samples selected shall conform to 4.4.1 and shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 2.5 for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each tamper shall be subjected to the tests marked" X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.5 Inspection procedure. Prior to examination or tests, the tampers shall be lubricated with military oils and greases for use at the ambient temperature of the inspection site (see 3.9).

4.5.1 Examination. The tamper shall be examined as specified herein for the fol1owing defects:

Major

101. Material not as specified (see 3.3)
102. Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration (see 3.3.1).
103. Dissimilar metals not as specified (see 3.3.1.1).
104. Material and material finish or treatment information not available (see 3.3.1.2).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the tamper (see 3.3.1.3).
106. Engine not as specified (see 3.4).
107. Starter not as specified (see 3.4.1).
108. Fuel filters not provided (see 3.4.2).
109. Air filter not provided (see 3.4.3).
110. Muffler not as specified (see 3.4.4).
111. Fuel tank not as specified (see 3.4.5).
112. Tamping or ramming shoe not as specified (see 3.5).
113. Accessories, when required, not as specified (see 3.5.1).
114. Handles not as specified (see 3.6).
115. Drive system not as specified (see 3.7).
116. Compacting system not as specified (see 3.8)
117. Lubrication not as specified (see 3.9).
118. Maintenance characteristics not as specified (see 3.12).

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119. Health and safety features not as specified (see 3.15).
120. Weight and dimensions not as specified (see 3.17 and 3.18).
121. Workmanship not as specified (see 3.21).
122. Metal fabrication not as specified (see 3.21.1).
123. Bolted connections not as specified (see 3.21.2).

Minor

201. Special tools (if necessary) not provided (see 3.10).
202. Carrying case not specified (see 3.11).
203. Size of chest not as specified (see 3.11.1).
204. Blocking not as specified (see 3.11.2).
205. Paint not as specified (see 3.11.3).
206. Marking not as specified (see 3.11.4).
207. Loading not as specified (see 3.11.5).
208. Publication storage not as specified (see 3.11.6).
209. Warning and safety labels not as specified (see 3.15.1).
210. Technical publications not furnished (see 3.16).
211. Treatment and painting not as specified (see 3.19).
212. Identification and instruction plates not as specified (see 3.20).

4.5.2 Tests.

4.5.2.1 Test schedule.

TABLE I. Test Schedule.

First article 1	Individual 2	Tests 3	Test para. 4	Requirements paragraph 5
X		Performance	4.5.2.2	3.13
X	X	Run-in	4.5.2.3	3.13
X		Temperature	4.5.2.4	3.13
X		Humidity	4.5.2.5	3.13
X		Fuel test	4.5.2.6	3.13
X		Compaction frequency	4.5.2.7	3.13
X		Compaction force	4.5.2.8	3.13
X		Noise level	4.5.2.9	3.14

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4.5.2.2 Performance. Two tampers shall be operated either concurrently or nonconcurrently, on uncompacted soil at maximum governed engine speed for a period of not less than eight hours of compacting work per tamper. Nonconformance to the requirements of this specification or any failures as defined in 4.5.2.2.1 shall be cause for rejection of the first article tampers by the Government.

4.5.2.2.1 Failure definition. A failure is any malfunction, including but not limited to a malfunction or failure of any control, undesigned leakage of lubricant, leakage of fuel, loosening of any fasteners, or engine failure.

4.5.2.3 Run-in. Each tamper shall be subjected to a run-in of not less than 15 minutes during which all controls shall be adjusted to the proper settings and then operated. Any failure (see 4.5.2.2.1) shall constitute failure of this test.

4.5.2.4 Temperature. The tamper shall be operated at maximum governed engine speed for a period of not less than eight hours of compacting work (including downtime for normal servicing and refueling) at an ambient environmental temperature of not less than 110°F. Nonconformance to 3.13 or any failure (see 4.5.2.2.1) shall constitute failure of this test.

4.5.2.5 Humidity. The tamper shall be operated at maximum governed engine speed for a period of not less than eight hours of compacting work (including downtime for normal servicing and refueling) at an ambient environmental relative humidity of not less than 100 percent. Nonconformance to 3.13 or any failure (see 4.5.2.2.1) shall constitute failure of this test.

4.5.2.6 Fuel test. The tamper shall be operated at full engine throttle for a period of not less than one hour of compacting work at ambient environmental conditions. Nonconformance to 3.13 or any failure (see 4.5.2.2.1) shall constitute failure of this test.

4.5.2.7 Compacting frequency. The tamper shall be measured according to the Construction Industry Manufacturers Association, Light Equipment Manufacturers Bureau Standard Number 1. Nonconformance to 3.13 or any failure (see 4.5.2.2.1) shall constitute failure of this test.

4.5.2.8 Compacting force. The tamper shall be measured according to the Construction Industry Manufacturers Association, Light Equipment Manufacturers Bureau Standard Number 1. Nonconformance to 3.13 or any failure (see 4.5.2.2.1) shall constitute failure of this test.

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4.5.2.9 Noise level test. Noise levels shall be measured conforming to MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-l474 provisions shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing and marking shall be examined to determine compliance with section 5 requirements.

4.6.1 Unit of product. For purposes of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2 Sampling. Sampling for examination shall conform to MIL-STD-105.

4.6.3 Examination. Samples selected shall conform to 4.6.2 and shall be examined for the following defects. AQL shall be 1.0 percent defective.

124.	Preservation not as specified.
125.	Packing not as specified.
126.	Marking not as specified.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation. Preservatives specified herein shall conform to the applicable specifications listed and shall be applied in accordance with MIL-P-116.

5.1.1.2 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated as follows:

5.1.1.2.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

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5.1.1.2.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-8-121, type I or 11, grade A, class 2. The material shall be secured in place with tape conforming to MIL-T-22085, type II.

5.1.1.3 Engine. The engine shall be preserved in accordance with the level A requirements of MIL-E-10062 as specified for type II, method I.

5.1.1.4 Tools. Tools shall be preserved as specified in for unprotected surfaces in 5.1.1.2.

5.1.1.5 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.6 Carrying case. When a carrying case is furnished, the tamper and all items furnished with the tamper shall be consolidated in the carrying case.

5.1.2 Level C. The complete tamper and all furnished items shall be preserved in a manner that will provide protection against deterioration and damage from shipment to the initial destination. This level shall conform to the contractor's standard practice.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 When a carrying case is furnished. The case containing the tamper and all furnished items shall be covered with a close-fitting housing consisting of two sides, two ends, and a top, constructed in accordance with PPP-B-636, V3c, style optional. The housing shall extend to the bottom of the case, but not cover the skids, and shall be secured in place with not less than two straps conforming to QQ-S-781, Class 1, type I or IV, finish A, size 0.50 inch, thickness optional. Edge protectors shall be provided under the top straps.

5.2.1.2 When a carrying case is not furnished. The tamper and all items furnished with the tamper shall be packed together in a close-fitting box conforming to PPP-P-601, overseas type, style optional. The contents shall be blocked, braced or anchored as applicable within the box to prevent movement or damage. Box closure and strapping shall be in accordance with the appendix to the box specification.

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5.2.2 Level C. Each tamper and all items furnished with the tamper, shall be packed for level C as specified in 5.1.1 for level A except the fiberboard box shall be class domestic, the plywood cleated box shall be domestic type and the strapping shall be finish A.

5.3 Marking. In addition to any special identification marking required by the contract or purchase order (see 6.2), each container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tamper is intended for use in compacting soil in restricted and confined areas.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DODISS applicable and exceptions thereto (see 2.1.1).
c. Time frame required for "submission of first article (see 3.2).
d. When a type I is required or when a type II is required (see 3.13f).
e. When accessories are required (see 3.5.1).
f. When a carrying case is not required (see 3.11).
g. When marking is other than as specified (see 3.11.4).
h. When a loading plan is not required (see 3.11.5).
i. When special tools and special equipment other than specified are allowed (see 3.12).
j. When final paint color is other than as specified (see 3.19 and 3.11.3).
k. Level of preservation and level of packing required (see 5.1 and 5.2).
l. Any special marking (see 5.3).

6.3 First article model. Any changes or deviations of production tampers from approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article model will not relieve the contractor of his obligation to furnish tampers conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instruction materials, illustrated parts lists, contractor's maintenance and operational manual to be furnished with each tamper.

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6.5 Provisioning. The contracting officer shall include provisioning requirements for repair parts, as necessary, and instruction on shipment of tampers.

6.6 Lubricants. MIL-STD-838 Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113 Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.7 Subject term (key word) listing.

Tamper, Gasoline Engine Powered
Tamper, General Backfill
Tamper, Hand Operated
Tamper, Pan Type Compactor (Vibro Plate)
Tamper, Ram Type
Tamper, Soil, Compaction
Tamper, Upright Type Compactor (Rammer)
Tamper, Vibratory

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD

Preparing Activity:
Army - ME

Review Activity:
Army - AT

Project 3895-0285

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